CHEMISTRY

A reusable, impurity-tolerant and noble metal–free catalyst for hydrocracking of waste polyolefins

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One-step conversion of low-purity polyolefins to value-added products without pretreatments represents a great opportunity for chemical recycling of waste plastics. However, additives, contaminants, and heteroatom-linking polymers tend to be incompatible with catalysts that break down polyolefins. Here, we disclose a reusable, noble metal–free and impurity-tolerant bifunctional catalyst, MoS_x-Hbeta, for hydroconversion of polyolefins into branched liquid alkanes under mild conditions. The catalyst works for a wide scope of polyolefins, including different kinds of high–molecular weight polyolefins, polyolefins mixed with various heteroatom-linking polymers, contaminated polyolefins, and postconsumer polyolefins with/without cleaning under 250°C and 20 to 30 bar H_2 in 6 to 12 hours. A 96% yield of small alkanes was successfully achieved even at a temperature as low as 180°C. These results demonstrate the great potentials of hydroconversion in practical use of waste plastics as a largely untapped carbon feedstock.

INTRODUCTION

Around 10 billion tons of plastics have been synthesized since the beginning of large-scale applications of plastics in the 1950s (*1*). However, only used plastics meeting stringent requirements of purity can be recycled mechanically (*2*). Furthermore, most of these recycled plastics cannot undergo more than one recycling process due to the downgrading properties of recycled products (Fig. 1A) (*3*, *4*). Consequently, almost all used plastics end up in landfill, incineration, and even leakage into the ecosystem, resulting in severe environmental burdens (*5*, *6*). To address this issue, circular economy using low-purity waste plastics as economical and abundant substitutes for petroleum in chemical production of monomers or other value-added small molecules [liquid alkanes, arenes, etc. (*7*, *8*)], also known as chemical recycling or upcycling of plastics (*9*), has been actively pursued. Although some recent industrial projects have emerged for catalytic pyrolysis or cracking of used plastics at high temperatures (450° to 900°C) (Fig. 1A) (*10*–*13*), efficient conversion of unsorted plastic mixtures, especially low-purity polyolefins that constitute most waste plastics (*1*), to desirable small molecules is still urgently needed. The major challenge lies in the development of economical and robust catalysts for selective breaking of relatively inert internal C—C backbones of real-world postconsumer polyolefins in the presence of various additives, contaminants, and disposed plastics containing heteroatoms under relatively mild conditions.

Hydroconversion of polyolefins, collectively referring to hydrocracking and hydrogenolysis (*14*), exhibits great potentials in selectively yielding high-value liquid products, lowering the reaction temperature, suppressing undesired coke formation, and reducing the number of refining steps (*15*). Nevertheless, several key issues are yet to be resolved. First, no previous hydroconversion catalysts can simultaneously handle various waste plastics. Additives, contaminants, mis-sorted polymers, or multilayer products found in polyolefin waste streams can deteriorate the hydroconversion substantially. In particular, no single catalyst capable of hydroconverting both polyolefins and polymers with heteroatom linkages has been successfully developed (*16*). Thorough pretreatments are necessary to purify the feedstock to avoid catalyst deactivation (*17*). High marginal cost and additional environmental pressure of sorting different kinds of plastics and improving the degree of cleanness post stringent requirements on the catalyst. Second, catalysts known for low-temperature hydroconversion (mainly 200° to 300°C) generally use noble metals (Fig. 1A) (*18*–*23*), leading to prohibitively high operating cost according to techno-economic analysis (*24*). High selectivity to form low-value methane due to the rapid breakage of terminal C—C bond is another issue often encountered by relatively active catalysts using noble metals, especially Ru (*25*, *26*). Third, although several noble metal–free catalysts can transform high–molecular weight polyolefins into liquid products (*23*, *27*–*29*), their relatively harsh conditions and slow rates would significantly increase the capital cost as compared to that of catalytic cracking because of the necessity to use expensive alloys for pressure reactors that can tolerate hydrogen corrosions at high temperatures (>280°C) and pressures (>3.45 MPa) (*30*).

Herein, we report a noble metal–free, impurity-tolerant, highly dispersed MoS_x-Hbeta hydroconversion catalyst with tandem acidic cracking and hydrogenation in excellent synergy on both external and internal reaction sites of the zeolite. Efficient hydrocracking of polyolefins to a 96.6% yield of high-value liquid products and gas products at reaction temperature as low as 180°C was achieved. Even high–molecular weight, linear low-density polyethylene (LLDPE, M_w = 300 kDa), a relatively challenging substrate, was successfully converted to a 97.1% yield of high-value liquid hydrocarbons in a relatively short time (6 hours) under relatively mild conditions (250°C and 2 MPa H_2). The catalyst shows an excellent tolerance of various common additives and salt contaminants as well as an unprecedentedly wide scope of frequently used plastics. Both clean postconsumer polyolefins and their mixtures with

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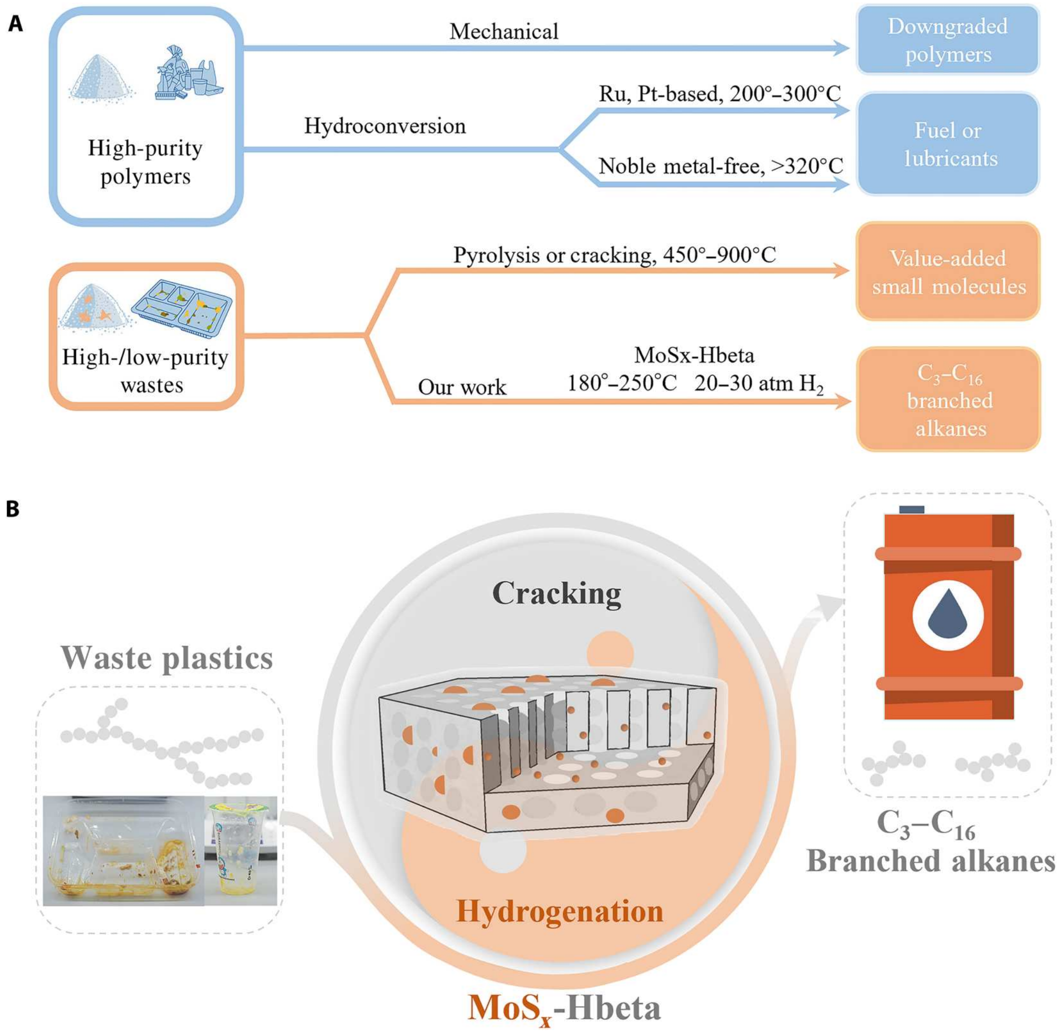

Fig. 1. Plastics recycling routes. (**A**) Schematic summary of current physical and chemical recycling routes of waste plastics. (**B**) Illustration of our hydrocracking method.

heteroatom-linked polymers, such as LDPE, LLDPE, high-density PE (HDPE), polystyrene (PS), polyvinyl chloride (PVC), polyurethane (PU), polycarbonate (PC), and polylactic acid (PLA), can be successfully transformed to high-value small molecules at relatively mild conditions. Such remarkable catalytic robustness allows for direct hydroconversion of real-world, contaminated post-consumer polyolefins to selectively form liquid products without any precleaning steps (Fig. 1B).

RESULTS AND DISCUSSION

We first studied the possibility of developing a noble metal–free bifunctional hydrocracking catalyst by a combination of zeolites for catalytic cracking and molybdenum sulfides, a classical type of industrial hydrodesulfurization catalyst with extraordinary tolerance of heteroatoms (sulfur, nitrogen. etc.), for hydrogenation (*31*). Hydrodesulfurization catalysts and HZSM-5, a commonly used type of zeolite, have been tested before for hydrocracking of polyolefins, but no synergetic effect was observed, which might be attributed to the difficulty of introducing hydrodesulfurization catalysts into the small channels of HZSM-5 (*32*). Thus, we reason that deposition of vaporized Mo complex into Hbeta, another type of zeolite with larger channels and higher activity in catalytic cracking of polyolefins compared to HZSM-5 (*33*), should provide a good chance to afford a highly disperse hydrocracking catalyst with a desirable synergistic activity.

To test our hypothesis, LDPE with high molecular weight (M_w = 300 kDa) was used as the initial substrate for the hydrocracking. The catalytic activity was evaluated by the sum of gas product and liquid product yields, defined as nonsolid yield (NSY), under the conditions of 200°C, 3 MPa H_2, 16 hours, and a polymer/catalyst ratio of 10 by weight. As shown in Fig. 2A, MoS_2 did not show any hydrocracking activity by itself, and Hbeta zeolite alone exhibited poor hydrocracking activity with only a 25% NSY. A simple mechanical blending of Hbeta with MoS_2 led to a slightly increased activity in hydrocracking (35% NSY). Encouragingly, a highly dispersed MoS_x-Hbeta catalyst, prepared from chemical vapor deposition of $Mo(CO)_6$ followed by sulfidation, substantially increased the NSY to 74.3% with an 81.1% selectivity on a carbon basis for C_5–C_{12} gasoline products. If a low–molecular weight PE (M_w = 4 kDa) was

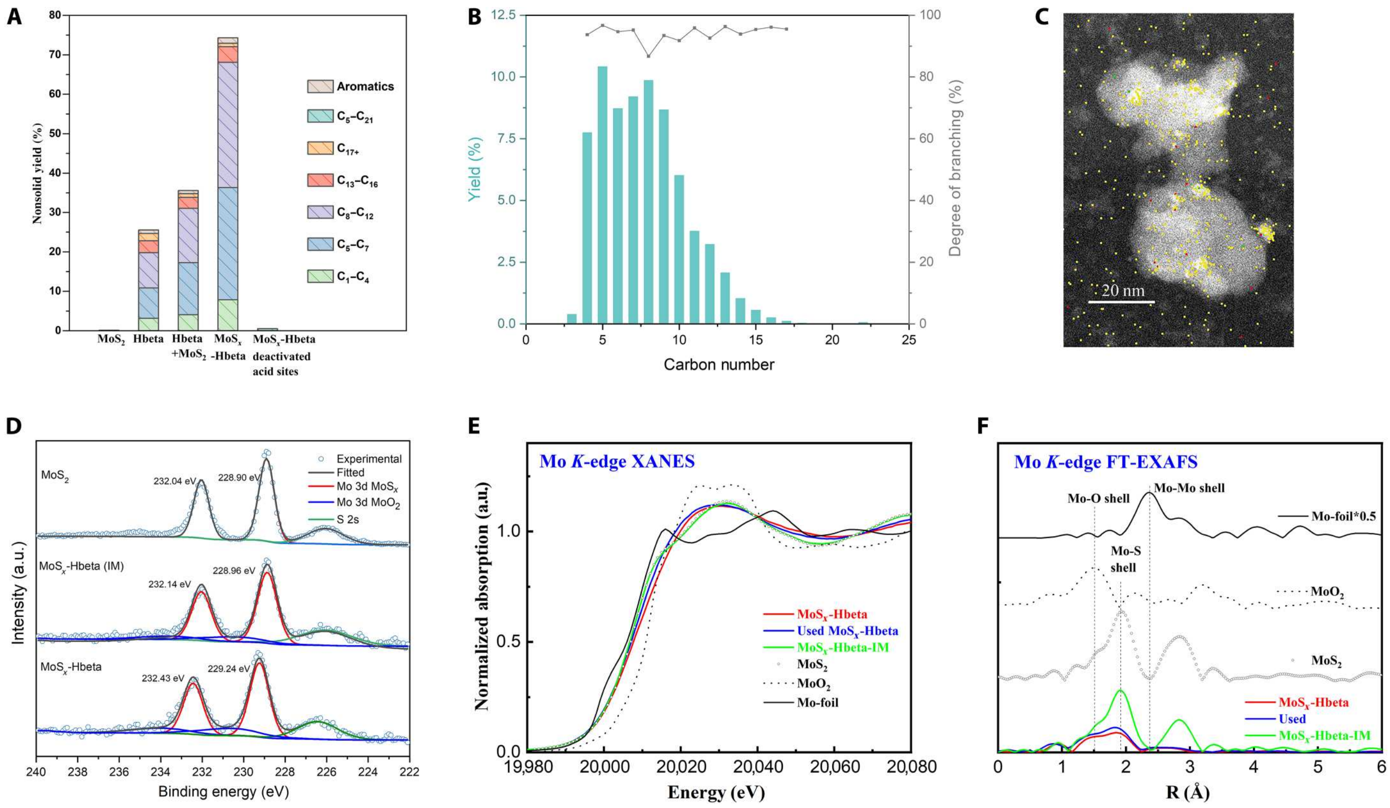

Fig. 2. Characterizations of the catalyst. (**A**) Different catalysts (1 g of LDPE, M_w = 300 kDa, 100 mg of catalyst, 200°C, 3 MPa H_2, 16 hours). (**B**) Product distribution of LDPE hydrocracking with MoS_x-Hbeta (1 g of LDPE M_w = 300 kDa, 100 mg of catalyst, 200°C, 3 MPa H_2, 16 hours). (**C**) Energy-dispersive x-ray spectroscopy (EDS) elemental mapping images MoS_x-Hbeta. The color yellow originated from the combination of green (Mo) and red (S). Scale bar, 20 nm. (**D**) The fitted XPS spectra of Mo. (**E**) Mo K-edge XANES spectra. (**F**) k^2-weighted Fourier transforms of Mo K-edge EXAFS spectra for Fresh MoS_x-Hbeta, used MoS_x-Hbeta, and MoS_x-Hbeta-IM with Mo-foil, MoO_2, and MoS_2. a.u., arbitrary units.

used instead of the LDPE, an NSY of 96% was obtained even at a temperature as low as 180°C (table S5). This overall catalytic performance of both high– and low–molecular weight polymers is superior to most available Ru and Pt systems in the literature from the consideration of reaction temperature, catalyst loading, reaction time, selectivity for liquid products, and molecular weight of PE (table S8), strongly supporting the validity of our design hypothesis for a highly dispersed synergistic bifunctional hydrocracking catalyst.

We then analyzed the detailed product compositions to gain preliminary mechanistic insights. Compared to a hydrogenolysis catalyst, MoS_x-Hbeta functions as a hydrocracking catalyst and leads to branched alkanes instead of linear alkanes as the major products owing to the excellent ability of Hbeta zeolite to isomerize alkanes (Fig. 2B). Branched alkanes are more desirable fuels due to their higher octane numbers than linear alkanes (*34*). The formation of thermodynamically favored alkyl-aromatics can serve as an evidence for the existence of alkene intermediates in classical acidic cracking mechanism, but the minor aromatic selectivity also demonstrates the efficiency of hydrogenation steps (fig. S24). MoS_x-Hbeta generated almost no C_1–C_2 alkanes, further supporting the dominant role of a classical acidic cracking mechanism involving β-scission of the C─C bond that does not provide C_1–C_2 carbeniums due to their extremely high energy. Upon a neutralization of Brønsted acid sites of MoS_x-Hbeta by exchanging H^+ with Na^+ in 0.1 M NaOH solution over 10 min, the hydrocracking activity is almost completely suppressed (Fig. 2A), confirming the pivotal role of the acidic cracking mechanism in the present catalysis. The acidity of Hbeta zeolite was further probed by infrared spectroscopy of pyridine adsorption (PY-IR) (fig. S34 and table S4), and the almost unchanged amount of acidity suggested that the deposition of MoS_x would not decrease the acidity of Hbeta.

We next characterized MoS_x-Hbeta by various tools to gain further mechanistic insights. X-ray powder diffraction (XRD) showed that the structural integrity of Hbeta was not destroyed upon the introduction of MoS_x (fig. S14). The absence of diffractions associated with MoS_x suggested that the size of MoS_x species should be below the detection limit of XRD. Further characterizations via high-resolution transmission electron microscope (HR-TEM), high-angle annular dark-field scanning TEM (HAADF-STEM), and corresponding element mapping for Mo and S indeed confirmed the presence of highly dispersed MoS_x nanoparticles (2 to 5 nm in size) (Fig. 2C and figs. S29 to S31). Spatial distribution of the MoS_x nanoparticles was further explored by adsorption and desorption experiments. The external surface area decreased after the incorporation of MoS_x, again indicating the existence of the nanoparticles outside the channels of Hbeta (fig. S15 and table S6). Meanwhile, the micropore volume decreased by 12% with only minor changes in pore size distribution after loading MoS_x, suggesting that the incorporation of nanoparticles into the channels as designed should not damage the pore structure.

The spatial distribution of the MoS_x nanoparticles suggests that the hydrocracking might occur at both the external surface and the inner pores. However, the observation of a narrow bell-shaped (Fig. 2B and fig. S5) characteristic of pore-controlled product selectivity (*35*) suggests that the final products should be mainly formed inside the pores. The critical role of pores in the hydrocracking again confirmed the validity of our design hypothesis to disperse a molecular Mo precursor into channels via vapor deposition.

We next examined the chemical state and the coordination environment of the Mo species in MoS_x-Hbeta by x-ray photoelectron spectroscopy (XPS) and synchrotron x-ray absorption spectroscopy (Fig. 2, D to F, and fig. S33). X-ray absorption near-edge structure (XANES) spectra exhibit that the valance of deposited Mo species is close to MoS_2. In XPS, MoS_x-Hbeta shows peaks at 229.2 and 232.4 eV (Mo $3d_{5/2}$ and $3d_{3/2}$), which are slightly higher than those of standard MoS_2 (228.9 and 232.0 eV). These shifts strongly suggest the presence of a strong interaction between MoS_x and Hbeta featured with electron transfer from the former toward the latter. A detailed analysis of extended x-ray absorption fine structure (EXAFS) has revealed the unsaturated coordination number of MoS_x, which correlates well with the observation of an S/Mo ratio slightly smaller than 2 in XPS (fig. S33). In addition, the lack of a secondary Mo-Mo signal in the secondary coordination sphere (table S7) confirms the high dispersity of MoS_x nanoparticles, as observed in HR-TEM and HAADF-STEM. Furthermore, the coordination of O to the Mo center, which might come from the support, demonstrates the strong interaction between the deposited MoS_x and the support.

In comparison, we synthesized a control sample (MoS_x-Hbeta-IM) with an Mo loading similar to MoS_x-Hbeta by replacing the vapor deposition of $Mo(CO)_6$ with a solution impregnation of $(NH_4)_2MoS_4$. The control sample also shows similar XPS signal (229.0 and 232.1 eV) and XANES signal with MoS_2. As to the EXAFS analysis, although the coordination number is smaller than standard due to the activation in H_2/N_2, still, no Mo-O signal can be detected, suggesting no interaction between Mo and Hbeta support in MoS_x-Hbeta-IM. MoS_x-Hbeta-IM also showed a catalytic activity lower than MoS_x-Hbeta but higher than the mechanical blending of MoS_2 and Hbeta (table S5). These results suggest that MoS_x-Hbeta-IM should have a weaker interaction with the support than MoS_x-Hbeta, and a stronger interaction appeared beneficial to the desired catalytic activity. Given the fact that the hydrocracking rate is positively dependent on the pressure of H_2 (Fig. 3D), there should be some process involving H_2 before the rate-determining step in the catalytic cycle. Such process might be facilitated by the relatively strong interaction between MoS_x and Hbeta.

We further evaluated the effect of reaction temperature, time, pressure, and the loading of catalyst for MoS_x-Hbeta. NSY increased as time, temperature, pressure, and the loading of catalyst increased (Fig. 3, A to D). Notably, an NSY of 82.4% was achieved with a polymer/catalyst ratio of 20 (Fig. 3A). The NSY was successfully

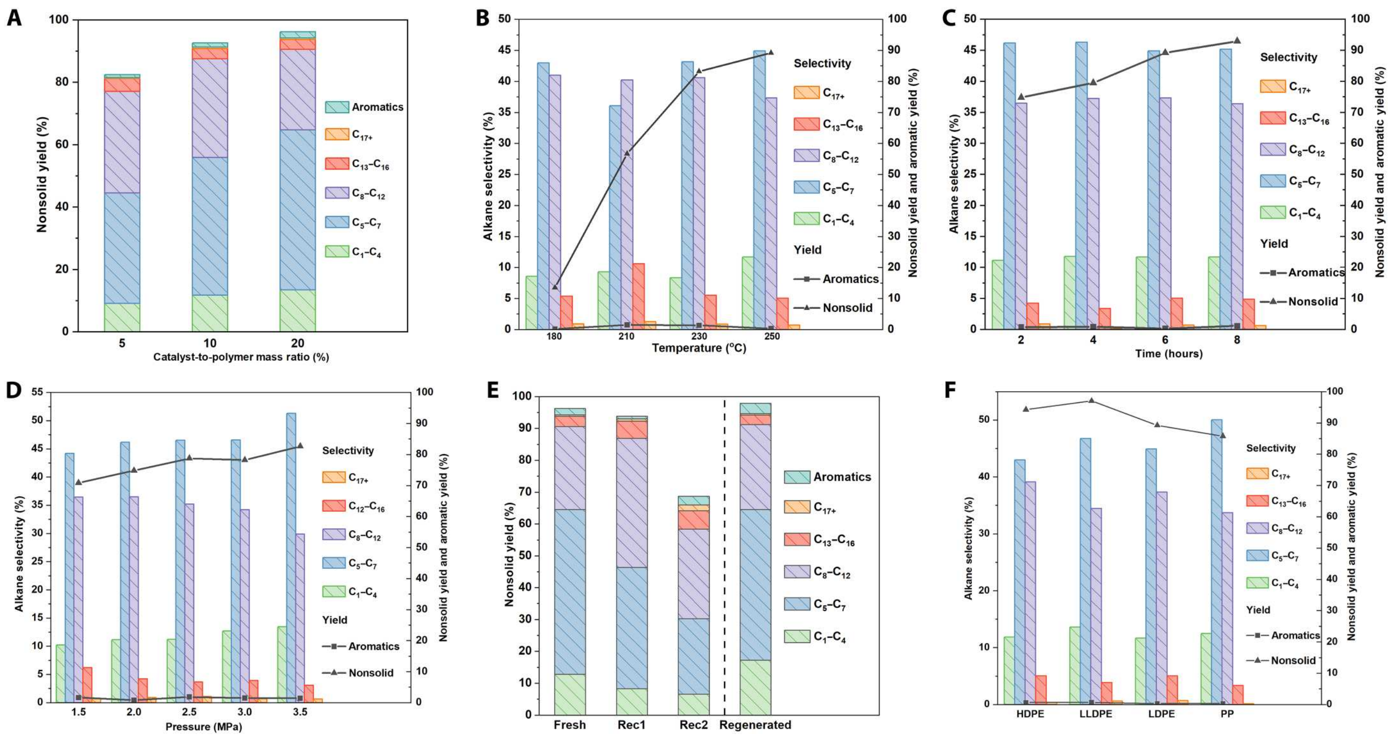

Fig. 3. Catalytic hydrocracking performance. (**A**) Influence of catalyst loading (1 g of LDPE, M_w = 300 kDa, 250°C, 6 hours, 2 MPa H_2). (**B**) Influence of temperature (1 g of LDPE, M_w = 300 kDa, 100 mg of catalyst, 6 hours, 2 MPa H_2). (**C**) Influence of time (1 g of LDPE, M_w = 300 kDa, 100 mg of catalyst, 250°C, 2 MPa H_2). (**D**) Influence of pressure (1 g of LDPE, M_w = 300 kDa, 100 mg of catalyst, 250°C, 2 hours). (**E**) Catalyst recyclability [catalyst/polymer ratio of 20% by weight (see the Supplementary Materials for more explanation), 250°C, 6 hours, 3 MPa H_2, catalyst regenerated, 8 hours]. (**F**) Catalytic hydrocracking with different polyolefins (1 g of polymer, 100 mg of catalyst, 250°C, 2 MPa H_2, 6 hours). These values were calculated on carbon basis.

improved to 89.2% by increasing temperature to 250°C compared to the 13.5% yield at 180°C (Fig. 3B). As time proceeds, the chain length shortened as revealed by differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA) (figs. S34 and S35), but the product selectivity remained virtually unchanged, suggesting that the pores may control the final product distribution (Fig. 3C). In general, higher NSYs were accompanied by higher selectivity for alkanes with lower carbon numbers, owing to the greater extent of hydrocracking as compared to systems with lower NSYs (Fig. 3, B and D). Owing to the shape selectivity of the zeolite, C_5–C_{12} alkanes were found to be the main liquid products (figs. S5 to S8), while a shift of production distribution from heavy alkanes (>C_7) toward light alkanes (≤C_7) was observed upon temperature and pressure increase, showing the potential to customize products as demanded by using the present catalyst.

According to techno-economic analysis, a combination of catalyst recyclability and low catalyst loading is important to reduce the cost for chemical recycling of waste polyolefins (*24*). Thus, a catalyst recycling experiment was also conducted for MoS_x-Hbeta. The NSY only decreased slightly in the first round of recycling (Fig. 3E). The coordination environment remained in EXAFS (table S7), suggesting the robustness of the interaction support and MoS_x. However, after a second-round catalyst reuse, the NSY decreased from the initial 93.8 to 67.8% presumably due to the blockage of catalytic channels by solid residues and cokes. Delightfully, the catalyst can be successfully regenerated upon calcination at 500°C in air followed by resulfuration process, leading to an NSY of 97%. Characterization of the catalyst after 10 hours' hydrocracking at 250°C and the regenerated catalyst after being used three times revealed no obvious aggregation of MoS_x particles, according to the HAADF-STEM (figs. S30 and S31), consistent with the robustness of MoS_x-Hbeta observed in the catalyst-recycling tests.

We next probed the applicability of MoS_x-Hbeta to different types of polyolefins with the same molecular weight (M_w = 300 kDa; Fig. 3F and figs. S9 and S10). Conversions of all commonly used aliphatic polyolefins, including HDPE, LDPE, LLDPE, and polypropylene (PP), into liquid products with high yields (85.7 to 97.2% at 250°C and 56.0 to 81.8% at 200°C) were successfully achieved. Compared to HDPE and LLDPE, PP and LDPE showed lower NSYs and lower selectivity of C_{8+} liquid. The branched structures of PP and LDPE presumably slow down the rate of the polymer chain to enter the channel of Hbeta (*36*, *37*), while the bond strength of tertiary C─C bonds is weaker than secondary C─C bonds (*38*), probably leading to lower activation barriers in C─C scissions. The combination of these two factors should account for the ostensibly counterintuitive combination of low NSYs and low selectivity of C_{8+} liquid in the hydrocracking of PP and LDPE.

Recent works have studied the applicability of catalyst in hydroconversion under scenarios including clean postconsumer plastics (*39*–*42*), different additives (*43*), and mixed polyolefins (*19*, *44*–*46*), but none of them were performed under different combinations of scenarios simultaneously. Herein, we first explored the application potential of MoS_x-Hbeta in chemical recycling of real-world waste plastics. Hydrocracking of commercial PE food wrap, PE pipettes, and PP melt-blown nonwoven surgical masks was tested initially (Fig. 4, A and B). These daily used plastics can all be successfully hydrocracked with good to excellent NSYs (53.7 to 94.1%) in 6 hours. The 53.7% NSY of PP melt-blown nonwovens surgical masks due to the interference of various possible additives can be further improved to 77.9% in 12 hours. Encouraged by these initial successes, we next investigated the robustness of MoS_x-Hbeta in the hydrocracking of polyolefins with various purposefully added impurities. The catalytic hydrocracking ability of MoS_x-Hbeta was tested for various polymers beyond PE and PP. Delightfully, MoS_x-Hbeta was found to be an efficient catalyst for hydroconversion of PS, PVC, thermoplastic PU (TPU), PC, and PLA at 250°C (table S5 and figs. S16 to S23), giving the first example of a single catalyst that can hydro-convert both polyolefins and polymers with heteroatom linkages. By comparison, it is well known that chlorine present in PVC can readily inactivate hydrogenolysis Ru catalysts due to its bonding to Ru (*21*, *47*). The resistance of both MoS_x and Hbeta to chlorine confers exceptional PVC hydrocracking ability to our catalyst, as evidenced by the fact that the addition of 5% PVC has almost no effect on the hydrocracking of LDPE (fig. S12). Moreover, the addition of 5 weight % (wt %) of different plastics as additive in hydrocracking of LDPE was explored as well (fig. S12). Except for PA6, most of the systems preserved moderate-to-excellent NSYs (66.1 to 82.3%) in hydrocracking of LDPE. The poor NSY of PA6 might be attributed to the amide group, as discussed below. An NSY of 72.0% in 6 hours and 81.4% in 12 hours (Fig. 4C and table S5) can be achieved for a mixture of LDPE, HDPE, PP, and PS prepared according to the percentage of polyolefins produced globally in 2015 (*1*). To such a mixture of polyolefins was deliberately added various polymers containing heteroatoms, including TPU, PC, PLA, and PVC, to mimic realistic mis-sorted plastics, which gave an NSY of 70.9% upon hydrocracking using MoS_x-Hbeta (table S5).

Subsequently, the influence of moisture, acidic atmosphere, the addition of frequently used plasticizers, and various inorganic salts was examined. No obvious decline in the catalytic hydrocracking activity was observed with a deliberate addition of 5% H_2O and 5% 1 M HCl solution. Various inorganic salts are also bystanders in this system. A 0.5 to 1% loading of typically used additives including antioxidant BPA, Irgafos 168, and slip agent erucylamide exhibits limited influence on the reactivity, suggesting that our catalyst can tolerate a certain amount of functional groups, for example, phenolic group, amide, and phosphate (Fig. 4D) (*43*). Despite the inspiring result, excessive additives may poison the catalyst. A 2% loading of erucylamide showed 61.2% NSY decrease (fig. S11), probably due to the competitive binding of functional groups with the acidic sites influencing the cracking and hydrogenation balance. Since the reaction proceeds via acidic cracking mechanism, the reactivity will be undoubtedly suppressed by strong basic contaminators, for example, piperidine and cyano group in 2,2′-Azobis(2-methylpropionitrile) (table S5). Moreover, additives that may collapse the zeolite structure, for example, NaOH solution or HF atmosphere, are also undoubtably unacceptable in our system.

Having successfully demonstrated the catalytic robustness of MoS_x-Hbeta in various artificial systems, we next tested realistic waste plastics as the hydrocracking substrates (Fig. 4A). The post-used disposable congee cup and lunch box without any cleaning pretreatments, which represent a large volume of daily kitchen plastic wastes, were directly hydrocracked under the present protocol, giving good-to-excellent NSYs of 88.1 and 54.3%, respectively. The NSY of lunch box can be further increased to 96.5% in 12 hours. All these results indicate that MoS_x-Hbeta is a promising noble

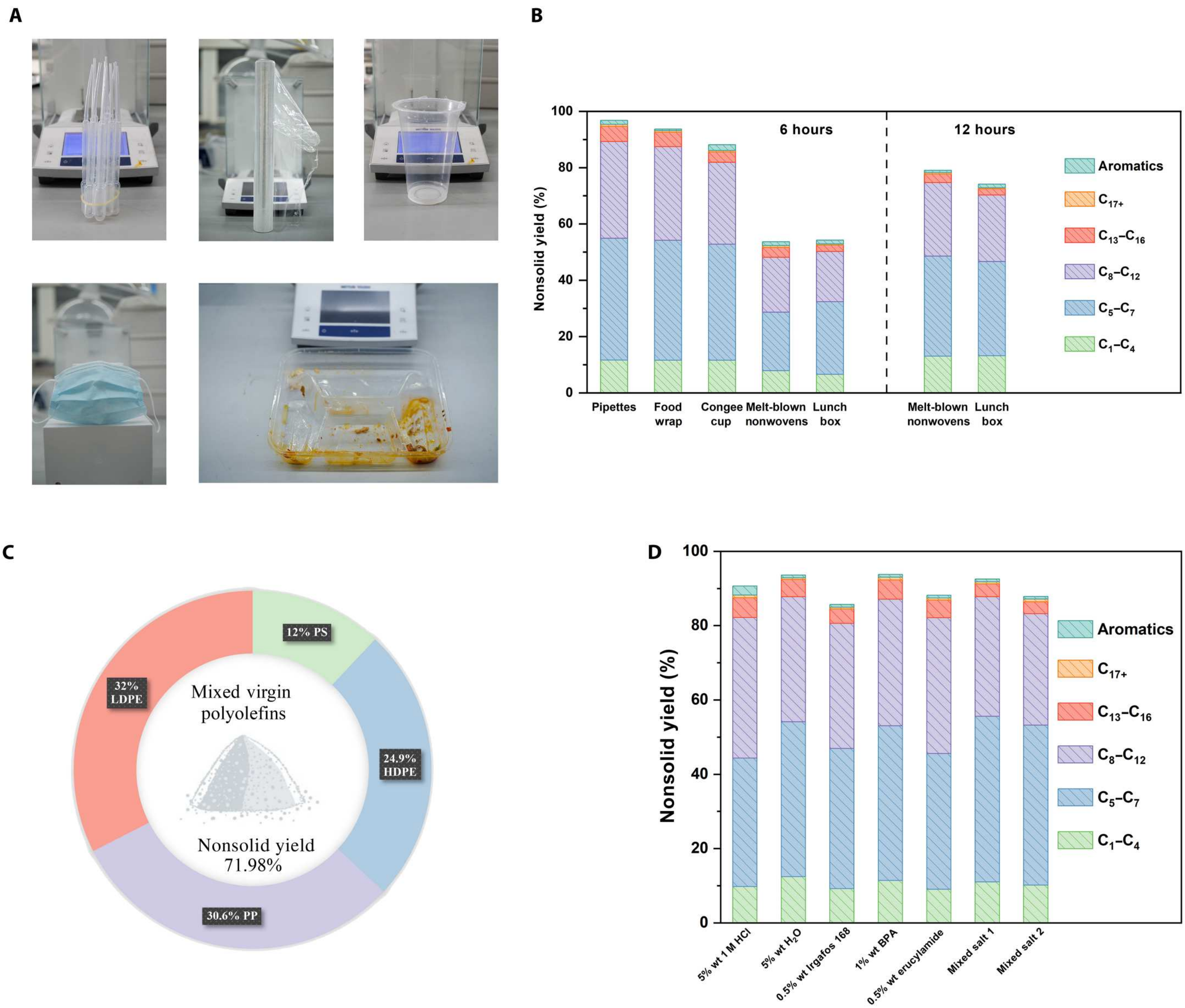

Fig. 4. Hydrocracking performance of real-world plastic systems. (**A**) Illustration of different post-consumer plastics used in (B). (**B**) Catalytic performance of real-world plastic systems (1 g of plastic, 100 mg of catalyst, 250°C, 2.5 MPa H_2, 6 or 12 hours). (**C**) Hydrocracking of mixed virgin polyolefins (1 g of mixed polyolefins, 100 mg of catalyst, 250°C, 2.5 MPa H_2, 6 hours). (**D**) Catalytic performance of 1 g of LDPE (M_w = 300 kDa) with different additives (mixed salts 1 including 50 mg of $CaCO_3$, 50 mg of $FeCl_3$, 50 mg of $MgSO_4$, and 50 mg of KBr; mixed salts 2 including 50 mg of Na_2CO_3, 50 mg of $PbSO_4$, 50 mg of $ZnCl_2$, and 50 mg of $CuBr_2$) (250°C, 2.5 MPa H_2, 6 hours, 100 mg of catalyst). These values were calculated on carbon basis.

metal–free, impurity-tolerant hydrocracking catalyst for efficient hydroconversion of various real-world post-consumer plastics.

In conclusion, we demonstrate that a noble metal–free, reusable, and contaminant-tolerant bifunctional catalyst MoS_x-Hbeta with a relatively strong metal support interaction can efficiently convert high–molecular weight polyolefins into branched liquid alkanes under mild conditions. The synergy between the C—C bond cracking ability of the acidic support and the excellent hydrogenation ability of the highly dispersed molybdenum sulfide led to highly efficient hydroconversion of polyolefins even at a temperature as low as 180°C. The wide substrate scope, including different kinds of plastics, mixed plastics, contaminated polyolefins, and post-consumer polyolefins with/without cleaning, clearly proves the validity of our synergistic designing principle in the development of novel catalysts for efficient chemical recycling/upcycling of real-world waste plastics.

MATERIALS AND METHODS

Materials

Most commercial reagents were purchased from Adamas, Macklin reagent, TCI, J&K chemical, and Alfa Aesar, and were used as received. NH_4beta zeolite was purchased from Alfa Aesar. $Mo(CO)_6$ was purchased from Energy Chemical. Different polymer powders including LDPE (M_w = 300 kDa, MB9500), PP (M_w = 300 kDa, H1500), LLDPE (M_w = 300 kDa, LL6210RQ), HDPE (M_w = 300 kDa, HMA-016), PP (M_w = 300 kDa, H1500), PS (M_w = 300 kDa, MN-1-310), PVC (M_w = 120 kDa, PBM-6), PS (M_w = 200 kDa, MN-1-310), PET (polyethylene terephthalate; M_w = 80 kDa, FB530), PU (M_w = 150 kDa, HT-8785), PA6 (M_w = 30 kDa, 1013B), acrylonitrile butadiene styrene (ABS) (M_w = 50 kDa, GP-22), PC (M_w = 45 kDa, 2407), PLA (M_w = 60 kDa, PBM-6), and PEwax (M_w = 3 kDa, AC-6A) were purchased from Dongguan WangDa Plastic Raw Material Company.

Catalyst preparation and regeneration

NH_4beta zeolite was first calcined under 450°C to obtain Hbeta, which was quickly transferred and stored in a glove box to avoid moisture adsorption. The Hbeta was mixed with $Mo(CO)_6$ in a round-bottom flask and then evacuated to ~3 to 4 mbar on the rotary evaporator for 1 hour at 35°C, followed by an activation process at 450°C for 3 hours in 100 standard cubic centimeters per minute (sccm) 8% H_2/N_2 flow. Subsequently, this activated Mo-Hbeta species was sulfided using 10 wt % S in the reactor with 1.5 MPa H_2 for 10 hours. After sulfidation, this catalyst was further activated in the 8% H_2/N_2 flow (100 sccm/min) in the tubular furnace to remove excessive physisorbed S. The prepared catalyst is labeled as MoS_x-Hbeta.

After hydrocracking, the catalyst was washed with dichloromethane and dried in the vacuum drying oven for next use. As for catalyst regeneration, the used catalyst will be first calcinated at 500°C for 3 hours in static air. Afterward, it underwent another sulfidation process and was reactivated using 8% H_2/N_2 atmosphere.

Catalyst characterization

The textural properties such as surface area (Brunauer-Emmett-Teller), external surface area (t-plot method), pore volume (t-plot method), and pore size distribution (nonlinear density functional theory method) of the samples were derived from N_2 adsorption-desorption measurements carried out at 77 K using a BELSORP-max II instrument. Before the measurements, ~150 to 200 mg of the powder sample was treated in vacuum at 120°C for 10 hours. XRD pattern spectra were acquired on a Bruker AXS D8 diffractometer using Cu-Kα radiation (λ = 1.54178 Å).

HR-TEM and HAADF-STEM were conducted on a JEM F200 field-emission TEM. ^{1}H nuclear magnetic resonance (NMR) spectra of the liquid product were recorded on a Bruker Avance 500 or 400 spectrometer at ambient temperature.

XPS data were determined by Thermo Fisher Scientific ESCALAB 250 Xi. H_2 temperature-programmed reduction (TPR) was recorded on Quantachrome Dynamic Flow Chemisorption and Reactivity Analyzer (Chemstar). Samples before any reduction were heated in Ar at 400°C for 30 min and then cooled down to room temperature. The TPR measurement was then carried out at a ramping rate of 10°C/min from 50° to 800°C in the flow of 10% H_2/Ar mixture. Transient gas was monitored with a thermal conductivity detector to measure H_2 consumption as a function of temperature.

^{13}C solid NMR was recorded on a Bruker Avance III HD 400WB (9.4 T) spectrometer equipped with 4- and 3.2-mm probes. The Larmor frequencies were 400.13 MHz for ^{1}H and 100.6 MHz for ^{13}C. Spectra were acquired at 10 kHz magic angle spinning. Direct ^{13}C excitation was achieved with a 3.84-μs 90° pulse. One hundred one kilohertz spinal 64 ^{1}H decoupling was applied during acquisition. A total of 256 to 1024 scans were acquired per spectrum.

TGA of fresh and spent catalysts was conducted on PerkinElmer TGA 8000 (TA Instruments) in the flow of the mixture of N_2 (20 ml/min) and O_2 (20 ml/min) in the range of 30° to 700°C with a 10°C/min heating rate. DSC of initial LDPE and solid residue was performed on PerkinElmer DSC 8000 (TA Instruments) in the flow of nitrogen (20 ml/min) with a heating rate of 10°C/min in 30° to 200°C range.

PY-IR experiments were performed on Bruker TENSOR II. The sample was first vacuum-pumped for 2 hours at 350°C. Then, the temperature was reduced to 40°C, and we recorded IR spectra as background signal. The sample was further treated with pyridine vapor at 150°C for 10 min. After saturation, the sample was vacuum-pumped to remove excessive pyridine. The temperature was increased with a 10°C/min rate to 350°C, and spectra were recorded at 150°, 250°, and 350°C.

The acquired EXAFS data were processed according to the standard procedures using the ATHENA module implemented in the IFEFFIT software packages. The k2-weighted EXAFS spectra were obtained by subtracting the post-edge background from the overall absorption and then normalizing with respect to the edge-jump step. Subsequently, k2-weighted χ (k) data of Mo K-edge were Fourier-transformed to real (R) space using a Hanning window (dk = 1.0 Å^{-1}). To obtain the quantitative structural parameters around central atoms, least-square curve parameter fitting was performed using the ARTEMIS module of IFEFFIT software packages.

The following equation was used in EXAFS fitting

$$\chi(k) = \sum_j \frac{N_j S_0^2 F_j(k)}{kR_j^2} \exp[-2k^2\sigma_j^2] \exp\left[\frac{-2R_j}{\lambda(k)}\right] \sin[2kR_j + \phi_j(k)]$$

N_j is the number of neighbors in the jth atomic shell. $S_0{}^2$ is the amplitude reduction factor. $F_j(k)$ is the effective curved-wave backscattering amplitude. R_j is the distance between the x-ray absorbing central atom and the atoms in the jth atomic shell (backscatterer). λ is the mean free path in angstroms. $\phi_j(k)$ is the phase shift (including the phase shift for each shell and the total central atom phase shift). σ_j is the Debye-Waller parameter of the jth atomic shell (variation of distances around the average R_j). The functions $F_j(k)$, λ, and $\phi_j(k)$ were calculated with FEFF8.4.

Reaction tests

The hydrocracking activity of prepared catalysts for different types of polymers was tested using a 10-ml autoclave reactor purchased from Anhui Kemi Machinery Technology Co. Ltd. We found that the overall conversion performance of a 40-ml reactor is higher than that of a 10-ml reactor under same reaction condition, possibly due to better temperature control. Consequently, we selected the 10-ml reactor for the experiment. Typically, 1 g of powder polymer was mechanically mixed with a certain amount of catalyst (50, 100, or 200 mg). The polymer catalyst mixture was loaded into the reactor together with a glass magnetic stirrer, subsequently sealed, and purged with hydrogen before being charged to the reaction pressure (2 to 3 MPa H_2). The set reaction temperature was reached within 30 min and then maintained for a specified time. After the reaction was completed, the reactor was air-cooled quickly to ambient temperature. As to the post-consumer plastics, they were cut into small pieces and tested similarly.

Note that we used a 20 wt % catalyst loading to prevent the catalyst loss in washing and in weighing, especially in the case that the catalyst has strong electrostatic effects when it is dry. However, if we increase the amount of LDPE to keep a 10 wt % catalyst loading in the 10-ml reactor, it will consume too much hydrogen, which will cause a significant high pressure drop and make it difficult to maintain the original pressure in the 10-ml reactor. Therefore, we ultimately chose a slightly higher catalyst loading to facilitate the cycling experiments. We acknowledged that a higher catalyst loading may lead to overestimation of the catalyst performance to some extent.

Product analysis

Gas products from the head space of the reaction were collected with a gas sampling bag at room temperature and analyzed with a Shimadzu gas chromatograph (GC) equipped with an AgilentHP-AL/S column and a flame ionization detector (FID). Quantification was conducted using standard calibration mixtures. Hydrocarbon products were extracted from the liquid-solid residuals using 4 ml of dichloromethane. The solution was analyzed for the selectivity in the solution on a Thermo Fisher Scientific Trace 1300 gas chromatograph equipped with a 60m DB-5 column and an FID. Representative raw GC figures are shown in figs. S3 and S4. Calibration of GC signals was conducted using normal alkane standard mixtures, and alkane isomers were assumed to have the same calibration factor as the corresponding normal alkanes (*39*). We did not figure out every peak in GCFID of liquid products due to the lack of calibrations of every kind of branched alkanes. As reported, the retention time of branched alkanes C_n is usually between the retention time of linear carbon C_{n-1} and C_n. To further minimize the possible peaking assigning error, we calculate the selectivity of alkanes with a carbon number between a certain range instead of every certain carbon number. Considering that most aromatic species are alkylaromatics according to the GC–mass spectrometry analysis and their total percentage are less than 4%, it is reasonable to assume that most aromatic compounds have a similar calibration factor, except benzene, whose calibration factor is quite different from others when using standard calibration mixtures. The calibration factor is derived from a standard mixture containing ethylbenzene and mesitylene. The remaining solid products were collected with excessive petroleum ether from the reactor. The PE solution was first evaporated at 45°C and then transferred to an 80°C vacuum drying oven overnight to gain the solid residue. The yield of solid residue (Y_{solid}) was evaluated gravimetrically. The yield of gas product alkanes (Y_{Gas}) was calculated using the following equation

$$Y_{\text{Gas}} = \frac{m_{\text{gas}}}{m_{\text{initial}}}$$

where m_{gas} is the mass of the total gas products (C_1–C_4), and m_{initial} is the mass of the initial polymer feedstocks.

The yield of the total liquid alkanes was calculated from

$$Y_{\text{Liquid}} = \frac{m_{\text{initial}} - m_{\text{gas}} - m_{\text{solid-residue}}}{m_{\text{initial}}}$$

The selectivity of product alkanes with *i* carbons (S_i) is evaluated from GCFID of nonsolid liquid products. The selectivity of certain liquid products was calculated from

$$S_i = \frac{Y_i}{\sum_{\text{non-solid}} Y_i}$$

Here, $\sum_{\text{non-solid}} Y_i$ corresponds to the yield of nonsolid products, namely, the combined yield of all gaseous and soluble liquid products. Note that the selectivity calculated from mass is the same as that calculated from carbon number.

Supplementary Materials

This PDF file includes:
Materials
Tables S1 to S8
Figs. S1 to S42
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A reusable, impurity-tolerant and noble metal–free catalyst for hydrocracking of waste polyolefins

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